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Biomedical Materials



PAPER

Biocompatibility evaluation of antioxidant cocktail loaded gelatin methacrylamide as bioink for extrusion-based 3D bioprinting

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Abstract

Three-dimensional (3D) liver bioprinting is a promising technique for creating 3D liver models that can be used for *in vitro* drug testing, hepatotoxicity studies, and transplantation. The functional performance of 3D bioprinted liver constructs are limited by the lack of cell–cell interactions, which calls for the creation of bioprinted tissue constructs with high cell densities. This study reports the fabrication of 3D bioprinted liver constructs using a novel photocrosslinkable gelatin methacrylamide (GelMA)-based bioink formulation. However, the formation of excess free radicals during photoinitiation poses a challenge, particularly during photocrosslinking of large constructs with high cell densities. Hence, we designed a bioink formulation comprising the base polymer GelMA loaded with an antioxidant cocktail containing vitamin C (L-ascorbic acid (AA)) and vitamin E (α -tocopherol (α -Toc)). We confirmed that the combination of antioxidants loaded in GelMA enhanced the ability to scavenge intracellular reactive oxygen species formed during photocrosslinking. The GelMA formulation was evaluated for biocompatibility *in vitro* and *in vivo*. These results demonstrated that the bioink had adequate rheological characteristics and was biocompatible. Furthermore, when compared to bioprinted constructs with lower cell density, high-density primary rat hepatocyte constructs demonstrated improved cell-cell interactions and liver-specific functions like albumin and urea secretion, which increased 5-fold and 2.5-fold, respectively.

1. Introduction

Liver is the second largest organ in human body that performs wide range of functions essential for normal body function. Therefore, a significant decline in liver function is detrimental to human health. Orthotropic liver transplantation is a tried and true treatment option for acute or end-stage liver diseases. However, scarcity of organ donors, high invasiveness and difficulties associated with life-long immune-suppressants has limited its benefits and increased the demand for engineering liver tissues [1].

To date various biofabrication strategies have been exploited to engineer functional liver that can

closely resemble their natural counterparts in terms of both architecture and functions. Three dimensional (3D) bioprinting is one such cutting-edge technology, which has the potential to precisely deposit biomaterials and cells in a layer-by-layer manner, to create heterogeneous tissue-mimetic constructs that promote cell–cell and cell-extracellular matrix (ECM) interactions in a 3D environment [2]. Liver is a complex organ, comprising of a variety of cell types at varying densities. Liver is organized into lobules and each lobule has well-arranged cellular architecture with majority of hepatocytes, that is arranged into hepatic chords separated by sinusoids [3]. Within the normal liver, the ECM comprises very limited compartment

which is less than 3% of the relative area on liver section [4]. Thus, liver has a high cellular density (1.3×10^8 cells per gram of tissue) and these densities need to be replicated during bio-printing to mimic the native tissue functionally and physiologically. The fabrication of high-density tissue constructs, on the other hand, is dependent on the bioprinting modality and bioink used.

Only very few methods for creating bioprinted tissues with high cell densities have been documented in the literature. Bedossa *et al* has reported the development a bioprinting approach to enable high-resolution transfer of spheroids into self-healing support hydrogels permits their patterning and fusing into high-cell density microtissues with predetermined spatial organization [5]. Daly *et al* on the other hand reported laser assisted bioprinting of engineered tissue with high cell density and successfully used this printing technology to print precisely miniaturized tissue like layouts with de novo high cell density and micro scale organization [6].

Extrusion bioprinters allow printing of tissue spheroids with 8000 cells in 300 μ m diameter and are hence preferred for bioprinting constructs with high cell density (e.g. cell aggregates) [7]. However, the efficacy of tissue fabrication by this printing modality is largely dependent on (i) physical and chemical properties of biomaterials used for cell printing to mimic tissue specific tissue ECM; (ii) shear-thinning properties to allow for fluidity within the printing nozzle for cell protection; (iii) cell-friendly cross-linking chemistry for post-printing stability and (iv) the ability to maintain the viability and functionality of embedded throughout the bioprinting process. This gave rise to the concept of creating structurally/functionally sophisticated biomaterial formulations, often referred to as bioinks, which are aqueous formulations of polymers or hydrogel precursors that contain cells and biological factors.

There exists a volume of literature on various kinds of hydrogels used for extrusion based bioprinting [8]. Among them, gelatin methacrylamide (GelMA) is widely used for bioprinting owing to its biocompatibility and superior printability. GelMA is readily photocrosslinkable in presence of photoinitiators like Irgacure 2959 and Lithium phenyl-2,4,6-trimethylbenzoylphosphine [9]. We have reported previously that during photo-crosslinking of GelMA, excess reactive oxygen species (ROSs) are generated within HepG2 cells embedded in GelMA and inclusion of a free radical scavenger (FRS) molecule, such as AA (vitamin C) in the GelMA bioink could nullify the adverse effect of excess ROS produced inside the cells [10]. The use of antioxidants as photoprotectants, either alone or in combination, has been extensively documented in the literature. For example, it is reported that a combination of vitamins C and E is superior for UV protection to skin than an

identical concentration of vitamin C or E alone. This is due to the fact that when vitamin C and vitamin E interact synergistically, they provide enhanced antioxidant protection. Vitamin E scavenges active free radicals during photo-crosslinking and gets oxidized to form vitamin E radicals. In the presence of vitamin C, vitamin E radicals get reduced and thus both work synergistically [11, 12].

In this study, we utilized this synergic mechanism to formulate a novel photo-protective bioink for the fabrication of bioprinted liver construct using high density (20×10^6 cells ml^{-1}) of metabolically active primary hepatocytes. The bioink formulation consisted of GelMA as base polymer and a cocktail of anti-oxidants vitamin C (L-AA) and vitamin E (α -Toc) along with a photoinitiator. The bioink was evaluated for its rheological characteristics, cell compatibility and biocompatibility. The bioink was used to fabricate liver construct and the construct was subjected to histo-morphometric and functional analysis.

2. Materials

Gelatin (Type A, porcine, 175 Bloom), Irgacure 2959, Fluorescein diacetate (FDA), propidium iodide (PI), cell counting kit (CCK-8), minimum essential medium (MEM), and Iscove's modified Dulbecco's medium (IMDM) were obtained from Sigma, India. Fetal bovine serum (FBS), trypsin-EDTA, and penicillin-streptomycin antibiotics were purchased from Gibco Invitrogen (India). Phosphate-buffered saline (PBS) was procured from TaKaRa (Shiga, Japan). The mouse subcutaneous fibroblast cell line (L-929) was procured from the American Type Culture Collection (USA) and human hepatocellular carcinoma (HepG2) cells were procured from the National Center for Cell Sciences (Pune, India).

2.1. Cell culture

Fibroblast (L-929) and hepatocyte (HepG2) cells were maintained in MEM supplemented with 10% FBS and 1% penicillin-streptomycin antibiotics in a carbon dioxide (CO_2) incubator set at 37 °C, 5% CO_2 , and 95% relative humidity.

Primary rat hepatocytes were isolated from male Wistar rats with prior approval from the Institutional Animal Ethics Committee (SCT/IAEC-406/JULY/2021/110) using a two-step perfusion method, as reported earlier [13]. Isolated hepatocytes were maintained in IMDM supplemented with 2% FBS and 1% penicillin-streptomycin antibiotics in a CO_2 incubator (New Brunswick, Germany) set at 37 °C, 5% CO_2 , and 95% relative humidity.

2.2. Synthesis of GelMA

GelMA was synthesized and characterized as previously reported [10, 14]. Gelatin was modified by reacting with methacrylic anhydride under optimum

temperature and pH. The reaction mixture was then dialyzed, lyophilized, and further sterilized by ethylene oxide gas at 37 °C.

2.3. Preparation of base GelMA

The base GelMA formulation was prepared by dissolving GelMA (5% w/v) in serum-free medium (MEM or IMDM) at 37 °C, which was later supplemented with 2-hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone (Irgacure 2959) at a final concentration of 0.5% (w/v) as photoinitiator. GelMA hydrogels were prepared by exposing 250 µl of the GelMA bioink formulation to 365 nm UV light at intensity of 8–12 mW cm⁻². For the cell culture experiments, GelMA bioink formulations were prepared under aseptic conditions.

2.4. Phototoxicity analysis

To estimate the effect of UV damage on GelMA encapsulated cells by photo-crosslinking, intracellular ROS (iROS) were measured by DCFDA-cellular ROS assay (Abcam ab113851), as previously reported [12]. HepG2 cells at a concentration of 9×10^5 cells were mixed with 600 µl of GelMA solution to complete the bioink. The bioink solution (200 µl) was transferred to 24 well plate and cross-linked by irradiation with UV light at 365 nm for 3 min to encapsulate the cells within the GelMA hydrogel. Similarly, cell monolayer in 24 well plate exposed to UV for 3 min were considered as positive controls, and normal cells without UV treatment were considered as negative controls. After UV exposure, cells from the GelMA hydrogel were harvested using collagenase A treatment. The control cells were treated with collagenase. The retrieved cell suspension was washed with PBS, resuspended in PBS (500 µl), and transferred into a black-clear 96 well plate and 2', 7'-dichlorofluorescein diacetate formed due to iROS activity was quantified by measuring the fluorescence at excitation/emission wavelengths of 485 nm/535 nm using a spectrofluorometer (Synergy H1, Biotek, USA). The percentage of ROS activity was calculated for the cells retrieved from GelMA and the positive control by normalizing with the negative control.

2.5. Photoprotective analysis of antioxidant cocktail loaded GelMA

An antioxidant cocktail was then added to the base GelMA bioink formulation. Two antioxidant vitamins, AA and α -Toc were used. The concentration of AA selected was 3.4 mM as previously reported [10]. The concentrations of α -Toc were 25 µM and 100 µM. A stock concentration of 1 M α -Toc was prepared in pure ethanol, diluted in PBS, and added to the GelMA formulation to obtain final concentration of 25 µM and 100 µM α -Toc. The UV protective property of GelMA-antioxidant formulations was analyzed by measuring iROS production by cells

encapsulated within the hydrogels, as discussed in the previous section. The percentage of ROS activity in cells within each GelMA-antioxidant formulation (3.4 mM AA, 25 µM α -Toc and 100 µM α -Toc) was calculated by normalizing with that of the base GelMA formulation.

2.6. Preparation of GelMA-antioxidant (GelMA-AO) formulation and evaluation

The GelMA-AO formulation was prepared at final concentrations of 5% GelMA, 3% gelatin, 0.5% irgacure 2959, 3.4 mM AA and 100 µM α -Toc. The Gelatin was added to impart thermosetting properties, enabling the bioink to be printed within the temperature range of 18 °C–20 °C. GelMA-AO hydrogel was prepared by crosslinking the solution under 365 nm UV light for 3 min.

2.6.1. In vitro cytotoxicity evaluation of hydrogel

The cytotoxicity of the GelMA-AO hydrogel was evaluated by direct contact and an extraction method based on ISO 10993-5.

The GelMA-AO formulation was preincubated at 37 °C and transferred to a 60 mm diameter sterile petri dish. The solution was crosslinked by exposing to 365 nm UV light for 3 min. The test sample for the direct contact test was prepared by cutting 4 mm discs from GelMA-AO hydrogel sheets. Ultra-high-molecular-weight polyethylene (UHMWPE) and organo-tin stabilized polyvinyl chloride were used as negative and positive cytotoxicity controls, respectively. Triplicate samples were placed on an L-929 monolayer and incubated at 37 °C for 24 h. The cells around the samples were examined under a microscope and cytotoxicity was graded based on the ISO 10993-5 standard.

Cytotoxicity was analyzed by exposing cells to the GelMA-AO hydrogel extract. L-929 cells were seeded in 96 well plate at a density of 1×10^5 cells/well and allowed to attain sub-confluency. Hydrogel discs with a thickness >1 mm was prepared using a Teflon mold. The extract was prepared by incubating the hydrogel discs in culture medium containing serum at an extraction ratio of 1.25 cm² ml⁻¹ in a shaking incubator at 37 °C for 24 h. The 100% extract was diluted with culture medium to obtain 50, 25 and 12.5%. GelMA-AO extract (100 µl) at each dilution was transferred to L-929 cells ($n = 8$) and incubated for 24 h. Cells in normal culture medium were considered as negative controls and cells treated with 1.3 mg ml⁻¹ phenol in culture medium were considered as positive cytotoxic controls. Cell morphology was examined microscopically for the response to extract treatment, and cell viability was determined by the 3-[4,5-dimethylthiazol-2-yl]-2,5 diphenyl tetrazolium bromide (MTT) assay. Cytotoxicity was determined by calculating the percentage cell viability expressed by cells treated with GelMA-AO extract

and the positive control after normalizing the values with that of the cell control.

2.6.2. HepG2 cell adhesion, viability and proliferation

The adhesion and morphology of HepG2 cells on the GelMA-AO hydrogel were studied using actin staining. HepG2 cells at a density of 3×10^5 cells/well were seeded on a GelMA-AO-coated cover glass placed on 12 well plate. Following culture for 72 h, culture medium was removed, and cells adhered to GelMA-AO were rinsed thrice with PBS and fixed using 4% paraformaldehyde for 30 min. The cells were permeabilized by treatment with 0.1% Triton-X 100 for 3 min and incubated with rhodamine-phalloidin (1:100 dilution in PBS) for 15 min. Cell nuclei were counterstained with Hoechst 33 258 ($0.05 \mu\text{g ml}^{-1}$ in PBS). The cells were observed under a fluorescence microscope (Leica DMI 6000, Switzerland) using filters N21 (red emission) and A (blue emission).

HepG2 cells at a density of 3×10^5 cells were encapsulated in 250 μl of GelMA-AO hydrogel by UV exposing for 3 min and cell viability was analyzed by live/dead staining immediately after photocrosslinking (0 h) and after day 1 (24 h). Cells were incubated in serum free medium containing 0.01% FDA and 0.005% PI. The cells were observed under a fluorescence microscope using I3 (green emission for live cells) and N21 (red emission for dead cells) filters.

Also, the proliferation of HepG2 cells (3×10^5 cells) encapsulated in 250 μl of GelMA-AO hydrogel was analyzed using cell counting kit—8 (CCK-8) on days 1 and 7. The culture medium was replaced with the CCK-8 medium (1:10 dilution in serum-free medium) and incubated for 2 h. The CCK-8 medium was collected in a fresh 96 well plate and the absorbance were measured at 450 nm using a multi-well plate reader (Biotek, USA). The percentage of cell activity was calculated by normalizing the absorbance value at day 7 with the average absorbance value obtained on day 1.

2.6.3. Rheology

Rheological properties of the GelMA-AO bioink formulation were analyzed using a rheometer (Anton Paar, Germany). Temperature sweep and viscosity variation of the pre-crosslinked GelMA-AO formulation were performed. Temperature sweep tests were conducted at varying temperatures (24°C – 8°C) to determine the physical gelation temperature of GelMA-AO formulation, and the viscosity variation of GelMA-AO solution at 20°C was measured at variable shear rate (1 – 1000 s^{-1}) using a parallel plate PP25, 0.5 mm measuring distance to study the shear-thinning behavior of GelMA-AO formulation. To investigate the UV photo-polymerization of GelMA-AO formulation, the sample were exposed to 365 nm UV light source Omnicure 1000 (Lumen Dynamix LDGI) at 20°C under a constant shear strain of 1%,

and the variation in storage modulus was recorded up to 200 s.

2.6.4. 3D printing

The printability of GelMA-AO formulation was primarily assessed by filament extrusion test followed by printing different patterns like square, triangle and zigzag to study the width size of the extruded filament, the ability of the bioink to print and maintain different angles. Similarly, a circle pattern was also printed to study the ability of bioink formulation to print curvature. A square grid multilayer pattern with internal perpendicular filaments with outer dimensions of $10 \times 10 \times 2 \text{ mm}$ was also printed to evaluate the shape fidelity of the bioink formulation. All the patterns were created using 3D BioCAD (regenHU, Switzerland) software and G-codes were generated to print at a speed of 10 mm s^{-1} at 20°C . Printing experiments were performed using extrusion print head of 3D printer (3D Discovery, RegenHU, Switzerland) with nozzle of size $410 \mu\text{m}$. Before initiating printing, the 5 ml cartridge containing the GelMA-AO solution was kept at 2°C – 8°C for 2–3 min to allow for physical gelation and the cartridge was loaded into a 3D printer and were printed based on the selected design. The printed multilayer constructs were cross-linked with 365 nm UV light for 3–5 min.

The surface morphology of 3D printed GelMA-AO constructs were analyzed by SEM (Model: Hitachi, S2400). The 3D printed samples were lyophilized, sputter coated with gold and observed under SEM. Images were captured and pore size was measured by image analysis using Fiji Image J software.

To study the stability of the 3D printed GelMA-AO hydrogel under physiological conditions, the constructs were incubated in MEM complete medium at 37°C in a CO_2 incubator for seven days. The initial weight (W_i) of the construct ($n = 4$) was measured on day 0, and the final weight (W_f) was measured after incubation on days (1, 3, and 7). The percentage of weight loss of the construct was calculated using the following equation:

$$\text{Weight loss (\%)} = \frac{W_i - W_f}{W_i} \times 100.$$

2.6.5. Subcutaneous implantation of 3D printed construct

The biocompatibility of the GelMA-AO hydrogel was evaluated in a rabbit model with prior approval from the Institutional Animal Ethics Committee of Bioneed India Private Limited (BIO-IAEC-4303). GelMA-AO constructs with a size of $10 \times 10 \times 2 \text{ mm}$ were 3D printed and used for subcutaneous implantation as per ISO 10993-6 biological evaluation of medical devices—tests for local effects after implantation. Clinically used collagen (ColoPlug, India) of a similar size to the test sample was used as a control. Adult male New Zealand White rabbits weighing 2.3–2.4 kg were subjected to surgical procedures

under general anesthesia by intramuscular injection using a mixture of xylazine (5 mg kg⁻¹) and ketamine (35 mg kg⁻¹).

The test item was implanted into the subcutaneous tissue on the right side by blunt dissection 25 mm apart from each implant, and the reference item pieces were implanted on the left side by blunt dissection 25 mm apart from each implant. A glass bead sterilizer was used to sterilize the surgical equipment. Subcutaneous pockets were made by blunt dissection, 25 mm from each site. The GelMA-AO hydrogel ($n = 4$) was implanted on the right side, and control collagen samples ($n = 4$) pre-soaked in sterile PBS were implanted on the left side. The skin incisions were closed using nonabsorbable suture materials. The animals were observed daily for mortality throughout the experimental period of 15 d. At the end of the observation period, animals were sacrificed by intravenous administration of sodium thiopentone. The implant sites, together with sufficient surrounding tissue, were collected and further processed for histopathological analysis. The tissue response at each implant site was determined, and the average score for the test and control groups was calculated. The average score for the control group was subtracted from the average score of the test group to obtain the histological evaluation score. The response to the test sample was graded as minimal or no-reaction (0.0–2.9), slight reaction (3.0–8.9), moderate reaction (9.0–15.0) or severe reaction (>15).

2.7. GelMA-AO bioink and optimization of high cell density 3D bioprinting

Liver construct was 3D bioprinted using GelMA-AO bioink (5% GelMA, 3% gelatin, 0.5% irgacure 2959, 3.4 mM AA and 100 μ M α -Toc and primary rat hepatocytes isolated from Wistar rats). Bioink with two different cell concentrations, 1×10^6 cells/construct (1 M) and 5×10^6 cells/construct (5 M) was prepared. Bioprinted constructs were stabilized under UV irradiation and cultured in IMDM + 2% FBS medium in a CO₂ incubator at 37 °C. Culture media were collected from the constructs and analyzed for albumin and urea production. Cell distribution inside the construct was assessed using hematoxylin and eosin (H&E) staining.

2.8. Functional evaluation of 3D bioprinted liver with high cell density

Culture media collected from bioprinted constructs (days 1, 3, and 5) were analyzed for urea and albumin secretion. The amount of urea released by the construct was estimated using a commercial urea assay kit (Biochain, Cat No: Z5030016) according to the manufacturer's instructions. The albumin secreted by the constructs was measured using a rat albumin ELISA Quantitation kit (Bethyl, Cat no: E111-125) according to the manufacturer's instructions.

The amount of albumin synthesized by the construct was determined from a standard curve obtained for rat albumin.

2.9. Statistical analysis

All experiments were performed in triplicate ($n = 3$). The results are presented as mean $\pm$ standard deviation, and significance was calculated using Student's *t*-test. Statistical significance was set at $P \leq 0.05$.

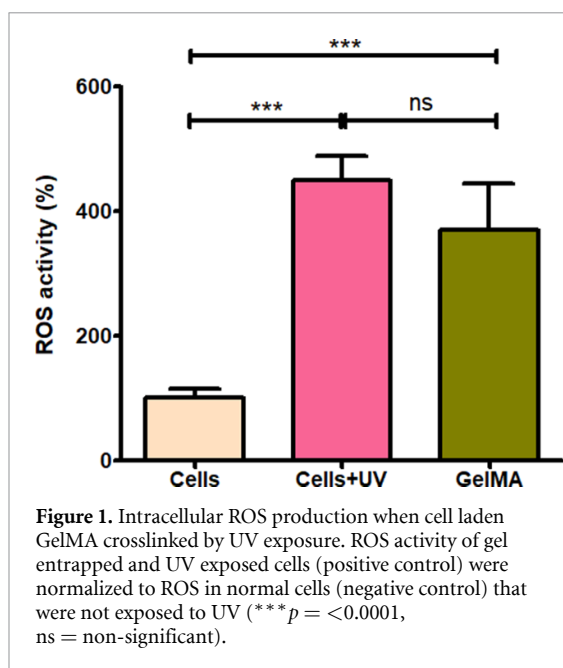
3. Results

3.1. Phototoxicity analysis in GelMA encapsulated cells

The formation of ROS in HepG2 cells exposed to UV light was measured to explain the phototoxicity. The iROS levels of HepG2 cells encapsulated in the base GelMA formulation were compared with those of normal cells (negative phototoxicity control) and cells exposed to UV (positive phototoxicity control). Figure 1 shows the ROS activity of the UV-exposed cells and cells encapsulated in the GelMA hydrogel. The UV exposed cells showed $450 \pm 16\%$ increase in iROS compared to normal cells without UV exposure. The GelMA entrapped cells also showed significantly higher ROS activity, reaching to $370 \pm 30\%$ compared to negative control.

3.2. Photoprotective effect of antioxidant loaded GelMA on cells

The photoprotective GelMA was formulated by optimizing the concentration of antioxidants and their subsequent effect on iROS production within encapsulated HepG2 cells. We have previously reported the standardized concentration of cytoplasmic protective agents by incorporating 3.4 mM AA into GelMA [10]. The effect of free radicals on the cell membrane may also affect cells. Hence, we evaluated two concentrations of α -Toc in GelMA (GelMA- α Toc 25 μ M and GelMA- α Toc 100 μ M) and compared the intra cellular ROS activity of cells encapsulated in the GelMA hydrogel (figure 2). The result indicates that when compared with conventional base GelMA formulation, the cells in GelMA-AA only $38 \pm 1\%$ of ROS activity in base GelMA. Similarly, cells in GelMA- α Toc 100 μ M also showed lower ROS activity of $57 \pm 3\%$ with respect to GelMA alone, whereas GelMA- α Toc 25 μ M did not show a significant difference. Hence, the antioxidant cocktail was formulated with 3.4 mM AA and 100 μ M α -Toc (GelMA-AA- α Toc 100 μ M) and analyzed for ROS activity. The results confirmed that the combination of AA and α -Toc had better efficiency in photoprotection, as evidenced by ROS activity of only $29 \pm 0.4\%$ compared to GelMA.



3.3. Bioink evaluation of GelMA-Antioxidant (GelMA-AO) formulation

3.3.1. In vitro cytotoxicity evaluation of hydrogel

In vitro cytotoxicity analysis of the GelMA-AO hydrogel based on ISO 10993-5 by direct contact with L-929 fibroblast cells did not show cytotoxic reactivity under or around the sample. Hence, it achieved a score of grade 0, which was considered non-cytotoxic (figure 3(a)). The negative control showed non-cytotoxic reactivity and the positive control showed severe cytotoxic reactivity.

The cytotoxicity due to the potential leachant from GelMA-AO hydrogel was analyzed by testing the extract method using L929 cells. After treatment with various dilutions of GelMA-AO extract, cells showed non-cytotoxicity with a cytotoxicity grade 0 based on ISO 10993-5. This was comparable to the morphology of cells cultured in the normal medium (cell control) (figure 3(b)). Cells treated with diluted phenol showed severe grade 4 cytotoxicity. To quantify the cytotoxicity, the MTT assay was performed soon after observation of cells at the end of the treatment time. The activity of cells treated with dilutions of GelMA-AO extract and phenol was normalized to that of the cell control. Cytotoxicity was calculated as a percentage of cell activity (figure 3(c)). The 100%, 50%, and 25% extracts of GelMA-AO showed mean cell activity of more than 100%, and there was no statistically significant difference compared to cell control. On the other hand, the cells treated with dilute phenol, which is considered a positive cytotoxic control, showed severe cytotoxicity with a Grade 4 score and <20% cell activity.

3.3.2. HepG2 cell adhesion, viability and proliferation

Staining of cytoskeletal structural proteins in HepG2 cells cultured on GelMA-AO hydrogel showed that

the hydrogel facilitated cell attachment. The cells showed a morphology on the gel surface and were evenly distributed (figures 4(a) and (a')). The viability of HepG2 cells encapsulated in the GelMA-AO hydrogel was analyzed by live-dead staining showed that cells were viable immediately after cross-linking, confirming the safe photocrosslinking process (figure 4(b)). The viability of HepG2 cells was analyzed after 24 h of encapsulation (figure 4(c)), confirming the suitability of GelMA-AO as a cyto-compatible hydrogel. The proliferation of HepG2 cells encapsulated in the GelMA-AO hydrogel analyzed by the CCK-8 assay is shown in figure 4(d). The cell activity measured on the 7th day after cell encapsulation was normalized to that measured on day 1. Assuming that the cells will multiply at least once in 7 d, the results substantiate the evidence for cell proliferation.

3.3.3. Rheology

The effects of temperature, shear rate, and time on the physical properties of the bioink were investigated by studying the rheological properties of GelMA-AO formulations. It was observed that the viscosity was inversely proportional to the temperature. A temperature sweep analysis of GelMA-AO formulation conducted in the range of 24 °C–8 °C is shown in figure 5(a). The gelation point is a sudden increase in viscosity with variation in temperature. GelMA-AO formulations showed a sudden increase in viscosity in the temperature range 18 °C–20 °C suggesting its gelation point. During extrusion, it is important that the bioink containing live cells be extruded through a small orifice of the nozzle without applying excessive shear to avoid the loss of cell viability. For this reason, it is necessary for the bioink to exhibit shear thinning behavior, which refers to the non-Newtonian behavior of fluids, so that the viscosity of the fluid decreases under shear. This shear-thinning behavior of the GelMA-AO formulation was evident from the viscosity curve shown in figure 5(b). The photo-crosslinking analysis under UV irradiation at a temperature above the gelation temperature of the GelMA-AO formulation is shown in figure 5(c), which shows the variation of storage modulus with respect to increasing UV exposure time from 50–200 s which clearly indicates that the GelMA-AO solution undergoes crosslinking.

3.3.4. 3D printing

Filament extrusion test shows the extrudability of GelMA-AO formulation under pressure which is an essential criterion for qualifying a bioink (figure 6(a)). Patterns printed using formulation suggest that the width of the extruded filament was same as that of the nozzle size (410 μ m inner diameter) used for printing with no shrinkage or swelling of the filament (figure 6(b)). The 3D printed multilayer

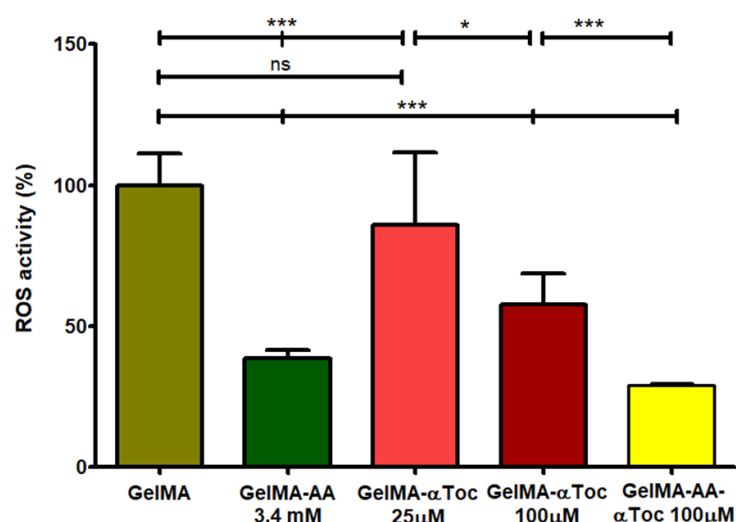


Figure 2. Photoprotective property of antioxidant loaded GelMA formulations. ROS activity in cells encapsulated in GelMA was considered as 100%. ROS activity in antioxidant loaded GelMA hydrogels were compared with GelMA. Combination of AA and α -Toc showed lesser ROS activity compared to all the formulations individually taken. ($n = 4$, *** $p < 0.001$, ns = non-significant).

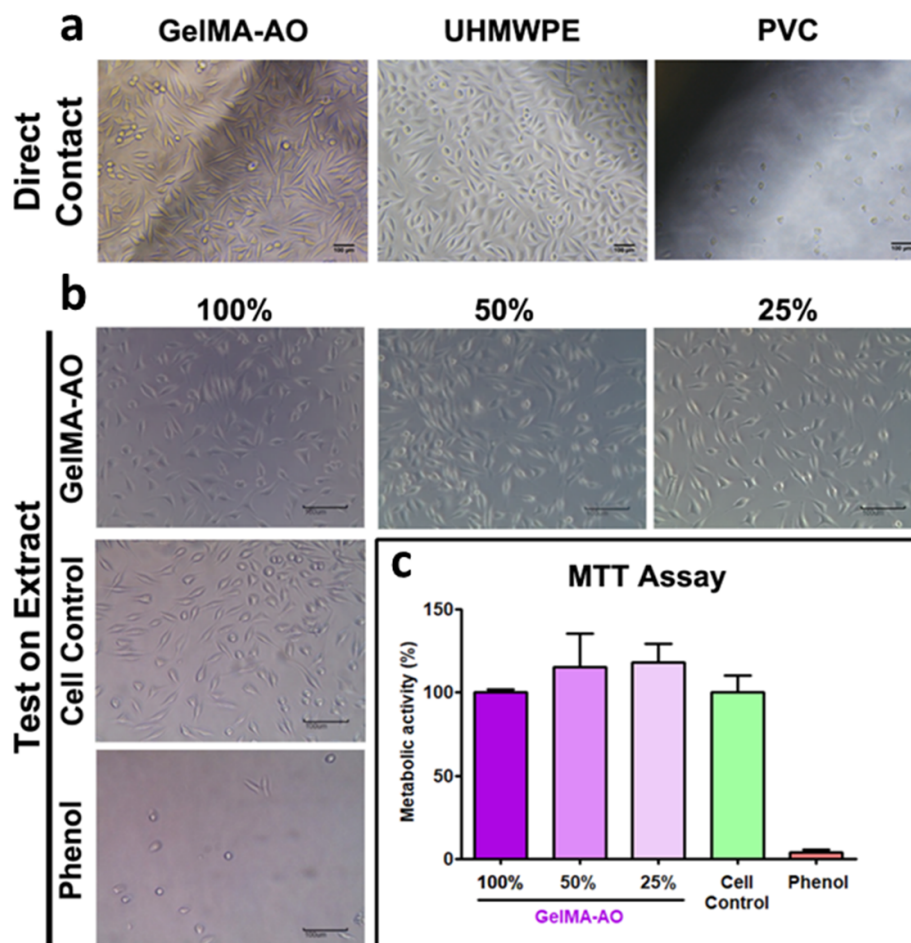


Figure 3. Cytotoxicity analysis of GelMA-AO hydrogel. (a) L929 cells after 24 h direct contact with GelMA-AO, negative control (UHMWPE) and positive control (1.3 mg ml^{-1} phenol). (b) Test on extract analysis showing L929 cells after 24 h treatment with dilutions of GelMA-AO extract, Cell control (culture medium) and dilute phenol. (c) Graph showing cell activity of cells determined by MTT assay at the end of test on extract.

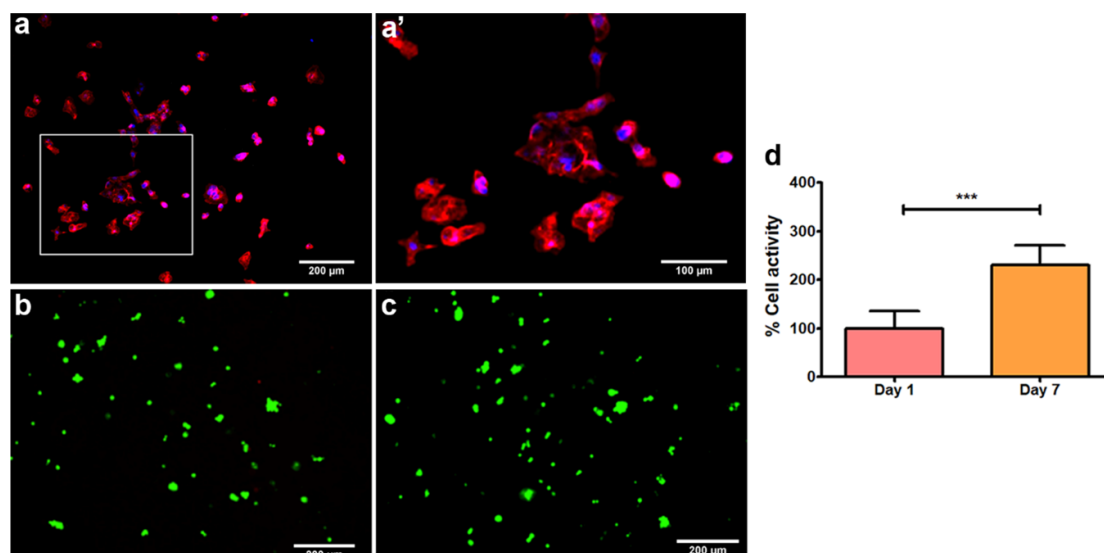


Figure 4. Cytoskeletal staining (red) of HepG2 cells adhered on GelMA-AO hydrogel. (a) Cells distributed over the surface showing adhesion. (a') Magnified area showing cell morphology and spreading. Cell nuclei were counter stained (blue) to identify the cells. Viability of HepG2 cells encapsulated in GelMA-AO hydrogel showing viability (green) at (b) 0 h and (c) 24 h of encapsulation. (d) Proliferation of HepG2 cells encapsulated in GelMA-AO bioink (***) $p = <0.0001$.

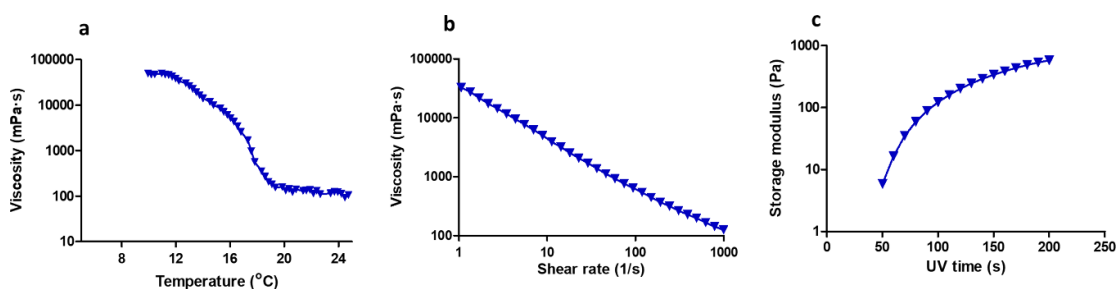


Figure 5. Rheological behavior of GelMA-AO bioink formulation (a) temperature sweep, (b) viscosity curve and (c) photocrosslinking.

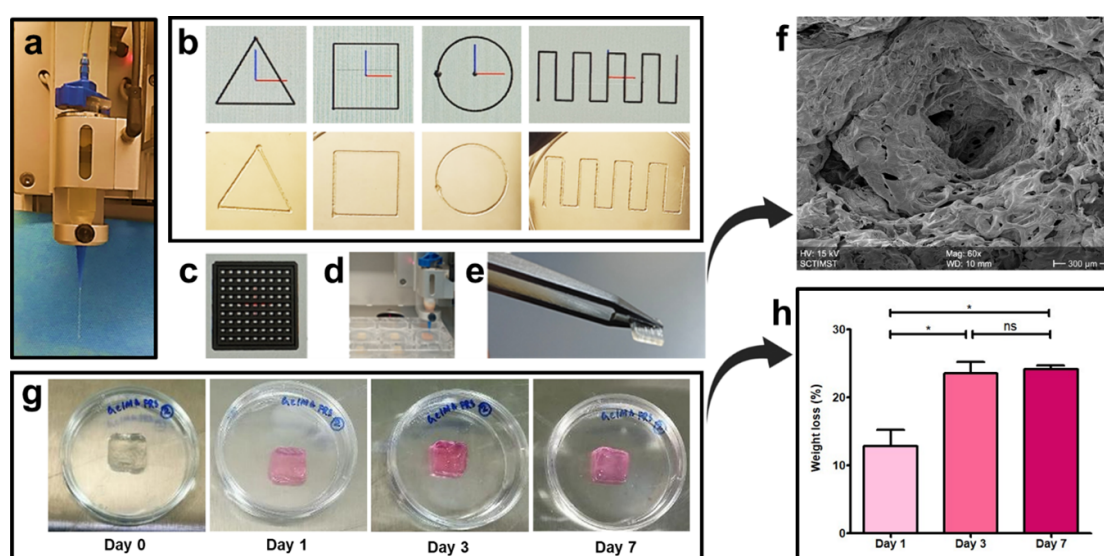


Figure 6. Three-dimensional printing of constructs with GelMA-AO formulation (a) The extrusion of GelMA as filament at appropriate pressure and temperature, (b) Printability of GelMA: first row represent various designs and second row shows corresponding printed GelMA, (c) 3D CAD design of multilayer construct, (d) 3D printing inside multiwell plate (e) 3D printed GelMA-AO construct, (f) SEM image of GelMA-AO construct showing porous nature, (g) 3D Printed GelMA-AO constructs incubated in culture medium for 0, 1, 3, and 7 d and (h) graph depicts the percentage weight loss of 3D printed constructs on day 1, 3 and 7 relative to day 0.

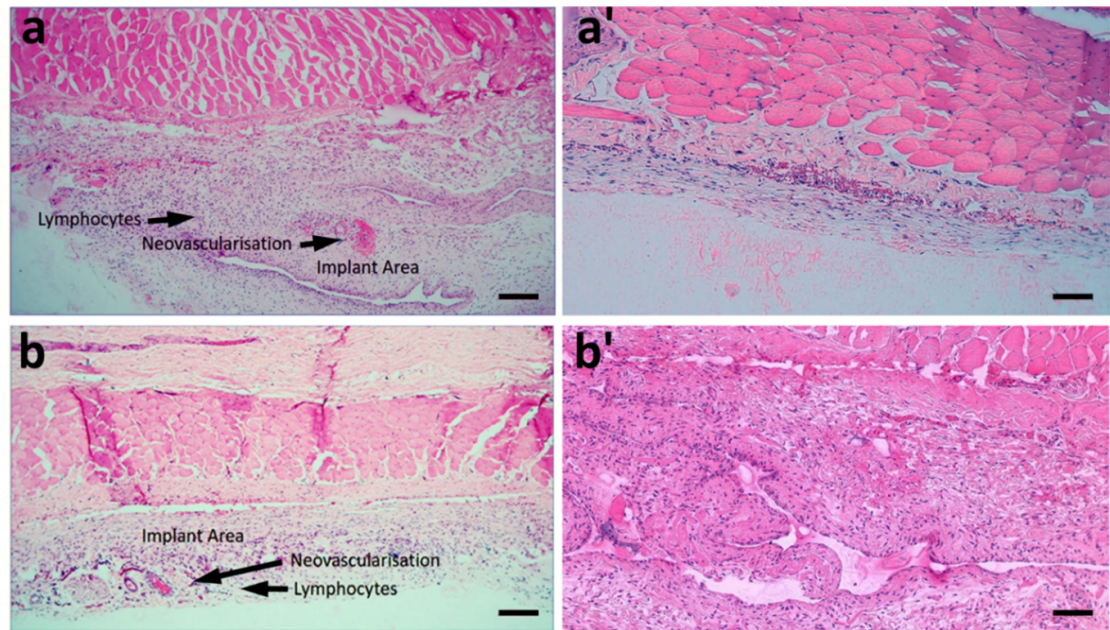


Figure 7. H&E staining of subcutaneous implanted 3D printed GelMA-AO construct. (a) and (a') showed the control sample under 5 $\times$ and 10 $\times$ magnification. Similarly, b and b' shows images of GelMA-AO at 5X and 10X magnification. Scale bar a&b = 200 μ m, a'&b' = 100 μ m.

square grid constructs also showed the same size as the CAD design (figures 6(c)–(e)).

3D printed GelMA-AO constructs observed under SEM showed filamentous and porous nature (figure 6(f)). The pore size measured by image analysis showed that the average pore size was 102 ± 20 μ m.

The stability of the 3D printed GelMA-AO construct under physiological conditions was analyzed based on weight loss over time. The constructs were stable in culture medium at 37 $^{\circ}$ C and maintained their shape over a period of 7 d (figure 6(g)). The quantitative analysis of weight loss depicted in figure 6(h) shows that there was a slight reduction of $9.6 \pm 6\%$ on day 1 and, reaching up to $20.2 \pm 6\%$ on day 3. On subsequent days, there were no further weight loss observed until day 7 ($20.7 \pm 6\%$).

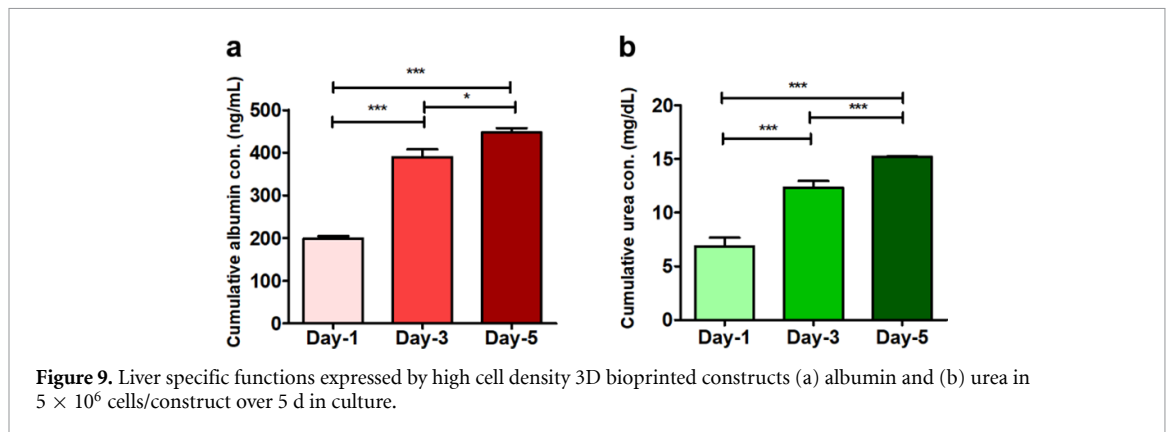
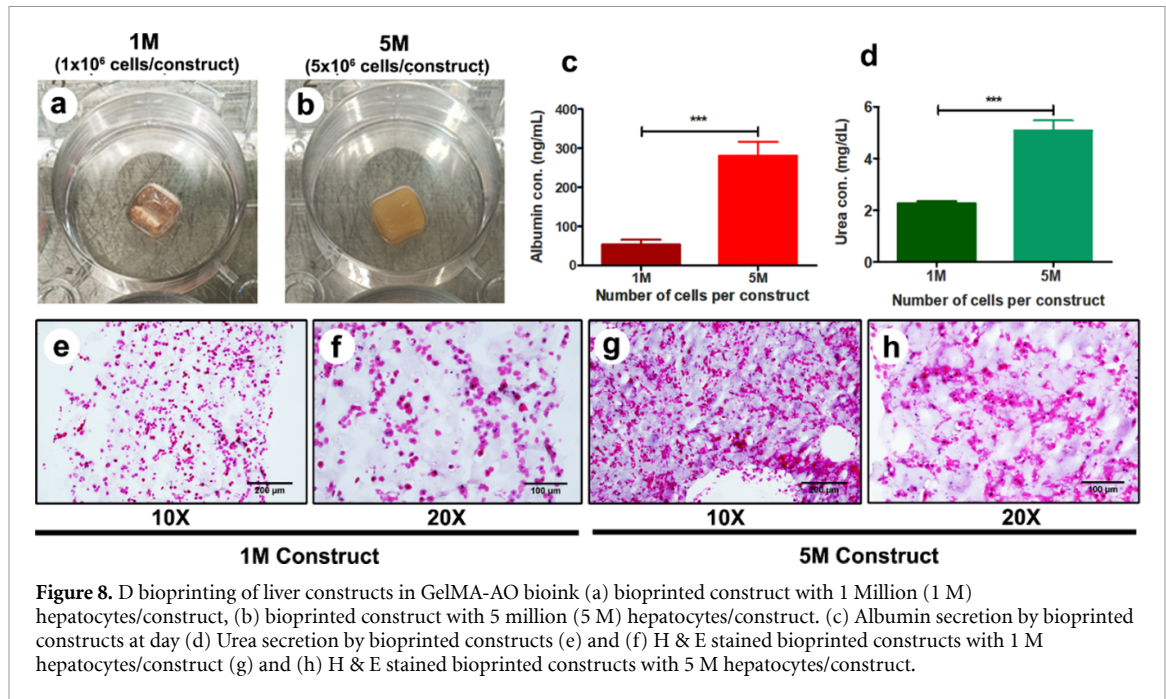
3.3.5. Subcutaneous implantation of 3D printed constructs

Subcutaneous implantation of 3D printed GelMA-AO construct did not show any signs of mortality or morbidity. All animals survived until the completion of the study. There were no treatment-related changes in body weight. There were no gross pathological observations during necropsy in any tissue, including at the implantation sites (subcutaneous tissue). Histological analysis of the tissue around the implant showed lymphocytic infiltration at implantation sites with a score of 1 in collagen control and scores of 1–2 in the GelMA-AO test item. Figures 7(a) and (a') shows macrophage infiltration around both the control and test sample implantation sites achieved a score of 1–2. The host tissue response in terms

of neovascularization in both control and GelMA-AO implantation sites is shown in figures 7(a) and (a'). Neovascularization was graded from 1 (minimal capillary proliferation, 1–3 buds) to 2 (group of 4–7 capillaries with supporting fibroblastic structures). The response at each implantation site was determined and the average score for the control and GelMA-AO sites was calculated. The average score for the control site was subtracted from the average score of the test site to obtain the histological evaluation score of the test item. The results showed that histological evaluation score for GelMA-AO was 0.6 (table 1: supplementary data). Thus, subcutaneous implantation of 3D Printed GelMA-AO hydrogel in a rabbit model resulted in minimal or no reaction to the subcutaneous tissue when evaluated at the end of 14 d.

3.4. Optimization of high cell density 3D bioprinting

The cell number for high cell density bioprinting was optimized at 4×10^6 cells ml^{-1} and 20×10^6 cells ml^{-1} concentrations of rat hepatocytes. A 3D construct of size $10 \times 10 \times 2$ mm requires approximately 250 μ l of bioink, which produces a construct with 1×10^6 cells (1 M) and 5×10^6 cells (5 M) as shown in figures 8(a) and (b). Albumin and urea synthesis are indicators of liver-specific functions in 3D bioprinted constructs. Figures 8(c) and (d) show that there was a significant increase in albumin and urea secretion by the 5×10^6 cells/construct compared to the 1×10^6 cells/construct. Albumin secretion was approximately 5-fold in 5 M constructs (281 ± 20 ng ml^{-1}) compared



to 1 M construct ($53 \pm 7.2 \text{ ng ml}^{-1}$), indicating maintenance of protein synthesis in high cell density constructs. The urea secretion in 5 M constructs ($5 \pm 0.2 \text{ mg dl}^{-1}$) was 2.5-fold compared to 1 M construct ($2.2 \pm 0.05 \text{ mg dl}^{-1}$). As the cell concentration increased, it was expected to have an enhanced function.

Histological analysis by H&E staining of 3D bioprinted cells laden with 1 M and 5 M liver constructs showed even cell distribution. The cells in the 1 M constructs were dispersed and spread out with minimal cell-cell contact (figures 8(e) and (f)). Interestingly, the 5 M construct showed more cell-cell interactions and orientations (figures 8(g) and (h)). Since the 5 M construct has a higher cell density, the hydrogel content was lower and more cell-cell interactions were evident. Based on the functional and histological evaluation, it can be deduced that 3D bioprinted constructs with 5×10^6 cells/construct outperformed those with 1×10^6 cells/construct, and

the 5 M construct was selected for bioprinting high cell density constructs for extended duration.

3.5. Functional evaluation of 3D bioprinted constructs with high cell density

The functional performance of the high cell density (5 M) 3D bioprinted constructs in exhibiting liver-specific characteristics was confirmed by analyzing albumin and urea secretion over an extended period of 5 d (figures 9(a) and (b)). Both albumin and urea secretion showed a linear increase over 5 d in culture.

4. Discussion

GelMA which forms covalently cross-linked hydrogels under UV light ($\lambda = 365 \text{ nm}$) exposure in the presence of a photoinitiator, is a widely used hydrogel in bioink formulations for 3D bioprinting. GelMA bioink has been studied for the fabrication of cardiac, skeletal, neural, bone, cartilage, and

skin tissues because of its excellent flow properties, biocompatibility, and controllable biodegradability [10, 15]. After 3D bioprinting, the cell-laden constructs were exposed to ultraviolet (UV) light. However, exposure to UV light may damage cells either by direct absorption of cellular DNA or indirectly via photosensitization reactions. During the photo-crosslinking of GelMA, a significant number of free radicals are produced during the radical-initiated crosslinking process. It is well established that exposure to UV light has a physiological impact by forming ROSs inside cells [16, 17]. In this study, we cross-linked GelMA by exposure to UV light at a wavelength of 365 nm and evaluated its phototoxicity to cells. We previously showed that the percentage of ROS formed in cells after UV crosslinking, results in extensive intracellular ROS production. The study also showed that the addition of 3.4 mM AA, a well-known FRS or antioxidant, to the GelMA bioink formulation offered photoprotection to the bioprinted parenchymal constructs with HepG2 cells by nullifying the UV radiation produced damage [10]. Similar experiment was also done in this study as control and shown in figure 1. This result was in line with that of our previous study, in which cells encapsulated with GelMA produced significantly higher ROS levels than cells not exposed to UV light. When we consider the 3D bioprinting of tissues with high metabolic activity, viability and functional performance must be maintained. In this study, we imparted a photoprotective property by formulating a cocktail of antioxidants such as AA and α -Toc. The role of these antioxidants in the bioink formulation was to enhance photoprotection during the crosslinking of liver constructs with metabolically active primary rat hepatocytes at high cell density. α -Toc is one of the extensively studied lipophilic molecules with antioxidant properties, and it has been reported that 100 μ M of α -Toc provides a protective effect against UVAI (340–400 nm)-induced DNA damage in keratinocytes *in vitro* [18]. Hence, α -Toc is considered a protective agent in topical sunscreens to prevent damage from sunlight. Based on previous reports, two concentrations of α -Toc (25 μ M and 100 μ M) were selected for this study and were included in the base GelMA formulation. Figure 2 shows that GelMA- α Toc 100 μ M significantly reduce the accumulation of intracellular ROS. We have previously reported that GelMA with 3.4 mM AA shows radical scavenging properties in bioinks. Here, we modified the formulation to enhance the photoprotective ability of GelMA for bioprinting liver constructs with high cell density. Analysis of ROS by cells encapsulated in GelMA loaded with a cocktail of AA (3.4 mM) and α -Toc (100 μ M) showed that the combination of antioxidants GelMA-AA- α Toc 100 μ M had a significant influence in reducing intracellular ROS compared to either antioxidant used alone. The synergistic interaction between AA and α -Toc was evident

in the outcome that resulted in better photoprotection when used in combination [12, 19].

In order to 3D bioprint liver construct with a maximal cell load and a minimum amount of matrix, 5% GelMA was used. It is also known that GelMA at a concentration of ≤ 5 w/v % is suitable for cell-laden constructs [20]. Low concentrations of GelMA affect 3D printability because of its poor viscosity and slow gelation rate. Addition of gelatin to 5% GelMA can improve the printability properties when it is used as a bioink. Therefore, we arrived at an optimized GelMA-AO formulation containing 5% GelMA, 3% gelatin, 3.4 mM AA, 100 μ M α -Toc and 0.5% irgacure.

To qualify the formulation for use as a bioink, it is essential to know its cytocompatibility [21, 22]. Cytocompatibility analysis of the GelMA-AO formulation was confirmed by cytotoxicity, cell adhesion, cell viability after encapsulation, and proliferation, as shown in figures 3 and 4. The actin filaments within the cells showed a direct evidence that the hydrogel promotes cell attachment. Bioink components influence its ability to be extruded in a consistent manner as well as maintenance of its structural integrity [23]. The rheological analysis of GelMA-AO revealed its thermoresponsive properties and enabled 3D printing at 18 °C–20 °C (figure 5(a)). Extrusion under pneumatic pressure might cause severe shear during bioprinting, which would affect cell viability [21]. Hence, one of the most important characteristics of bioinks is their shear-thinning behavior. The GelMA-AO formulation exhibited excellent shear-thinning behavior, as shown in figure 5(b). The addition of antioxidants may hamper crosslinking ability. Dynamic crosslinking analysis revealed an increase in the storage modulus with respect to the duration of UV exposure. Crosslinking within a duration of <200 s indicated that the antioxidants did not extensively affect the crosslinking property of the formulation (figure 5(c)).

Printability of a bioink is an important parameter to be considered while standardizing the bioink as the bioprinted product must mimic both cellular architecture and shape. The ability to extrude is a basic requirement for printability. A bioink formulation should be flowable and must be able to deposited precisely in terms of spatial, temporal, and volumetric control. Porosity of scaffold is important in tissue engineering as the pore size influences both nutrient and oxygen transfer for the survival, proliferation of cells and also in vascularization [24, 25]. It has been previously reported that pore size of 10% GelMA ranged 102 $\pm$ 20 μ m in diameter indicating suitable microenvironment for cell viability proliferation and function. The morphology of 3D printed GelMA-AO construct showed both macro and micro pores (figure 6(f)). Stability of the printed construct during *in vitro* maintenance under physiological conditions is another crucial factor. During the initial days, the observed reduction in weight could be due to

the release of gelatin from the construct [10]. Release of gelatin does not affect the shape or size of the constructs. Thus, it can be concluded that the 3D printed construct with GelMA-AO formulation was stable under physiological conditions (figures 6(g) and (h)).

The biocompatibility evaluation studies the local effects on the surrounding tissue (number and distribution of inflammatory cells, vascularity of fibrous capsule, granuloma, fatty infiltration, material debris, endothelialization, and presence of necrosis) following the implantation of biomaterials. We analyzed the biocompatibility of the GelMA-AO hydrogel in a rabbit model according to ISO 10993-6. When implanting hydrogels *in vivo*, a higher number of macrophages around the implant site can influence material degradation. Lymphocyte reactions to polymeric implants can also have negative effects and reduce biocompatibility [26]. In our experiment, collagen was used as the control material. When examined histologically (figure 7), the cell response in terms of macrophage infiltration at both control and test implantation sites was graded to a score of 1–2. Lymphocytic infiltration at the control implantation site achieved a score of 1 and the test implantation site achieved a score of 1–2. Vascular network formation at the implant site is critical for determining biocompatibility [27]. Neovascularization refers to the formation and growth of new blood vessels through several vascular processes, namely, vessel formation, sprouting, maturation, stabilization, remodeling, and specialization [28]. The tissue response in terms of neovascularization was graded as 1 (minimal capillary proliferation, focal, 1–3 buds) to 2 (group of 4–7 capillaries with supporting fibroblastic structures) at both the control and test item implantation sites. Thus, subcutaneous implantation of the 3D Printed GelMA-AO hydrogel can be considered to have minimal or no reaction to the subcutaneous tissue when evaluated at the end of 14 d.

The main requirement for 3D bioprinting of liver constructs is the fabrication of liver tissue that mimics the native tissue and exhibits liver-specific functions. Tissue engineering of liver tissues remains a challenge in developing functional constructs for clinical application and *in vitro* tissue models [29]. Some of the limitations of conventional tissue engineering can be overcome using 3D bioprinting. The advantage of 3D bioprinting is its ability to spatially distribute cells within the construct and to design the vasculature within the construct [30]. To achieve the highest level of functional performance in bioprinted constructs, it is necessary to mimic the cell arrangement and microenvironment of the native tissues. The liver tissue is made up of 70% parenchymal cells, 30% non-parenchymal cells, and fewer ECM components [31]. Hence, it would be preferable to consider the high cell density of hepatocytes and a low ECM compartment in biofabricated tissue. In this study, we designed a grid pattern of 1×1 cm

with an interfilament distance of 1 mm for efficient mass transfer. The bioink contained low (5% w/v) concentration of GelMA to accommodate high cell density. Dvir-Ginzberg *et al* reported high cell density bioprinting of liver constructs using alginate as a bioink [32]. Their study proved that a higher cell density of 5 million cells cm^{-3} maintained higher cell viability than those with low cell density. Here, we used GelMA-AO bioink to fabricate 5 million cells per construct with dimensions of $10 \times 10 \times 2$ mm, which is equivalent to 50 million cells cm^{-3} when extrapolated from a previous report. Histological analysis of cells in the 5 M cells/construct showed more cell-cell contact than the low cell density constructs. Albumin and urea synthesis analyzed 24 h after bioprinting showed 5-fold and 2.5-fold production, respectively, compared to the 1 M construct (figure 8). The 5 M constructs were further evaluated for an extended duration of 5 d, showing a linear increase in albumin and urea synthesis, substantiating the reason for improved functional performance in the high cell density construct, as shown in figure 9. Collectively, the results of this study provide evidence for the formulation of a photoprotective bioink for high cell density bioprinting of liver constructs.

5. Conclusions

This study reports the use of an antioxidant cocktail-loaded GelMA-based bioink with photoprotective properties for 3D bioprinting of liver constructs with high cell density. The results showed that the antioxidant cocktail system improved the radical scavenging properties of both antioxidants alone. Additionally, the 3D printed construct printed using the novel GelMA-AO formulation evaluated by subcutaneous implantation in a rabbit model confirmed its biocompatibility. Three-dimensional bioprinted constructs with a high density of hepatocyte cells showed improved functionality and cell organization compared to bioprinted constructs with low cell density. Our research proposes a potential method to overcome the phototoxicity of crosslinking bioprinted constructs using GelMA-AO bioink. The method and results presented herein will be useful for fabricating transplantable liver constructs that resemble native tissues.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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Author contributions

Conceptualization, P R Anil Kumar; Formal analysis, Anupama Sekar J; Funding acquisition, Shiny Velayudhan and P R Anil Kumar; Methodology, Anupama Sekar J and P R Anil Kumar; Project administration, P R Anil Kumar; Writing—original draft, Anupama Sekar J; Writing—review & editing, Shiny Velayudhan and P R Anil Kumar.

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